

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name	Software Design & Analysis	Course Code	CS3002
	Name		Exam Type	Quiz 5.v1
	Roll.No		Total Marks	/10
	Section:	5J	Weight	

Note : HAPPY LAST QUIZ ! , BEST OF LUCK FOR FINALS

Q1 : Improve the following code using an appropriate design pattern and give the solution code:

// Weapons: captainamerica, batman, superman and thor

```
weapon* createWeapon(heroName) {  
    switch(heroName) {  
        case CaptainAmerica: return new Shield();  
        case Batman: return new Sword();  
        case Thor: return new Hammer();  
        default: return new Hammer();  
    }  
}
```

Solution:

1. Weapon Interface

```
class Weapon {  
public:  
    virtual void use() = 0;  
    virtual ~Weapon() {}  
};
```

2. Concrete Weapons

```
class Shield : public Weapon {  
public:  
    void use() override { cout << "Using Shield\n"; }  
};
```

```
class Sword : public Weapon {  
public:
```

```

    void use() override { cout << "Using Sword\n"; }
};

class Hammer : public Weapon {
public:
    void use() override { cout << "Using Hammer\n"; }
};

```

3. Hero Abstract Class (Factory Method)

```

class Hero {
public:
    virtual Weapon* createWeapon() = 0; // Factory Method
    virtual ~Hero() {}
};

```

4. Concrete Hero Classes

```

class CaptainAmerica : public Hero {
public:
    Weapon* createWeapon() override {
        return new Shield();
    }
};

class Batman : public Hero {
public:
    Weapon* createWeapon() override {
        return new Sword();
    }
};

class Thor : public Hero {
public:
    Weapon* createWeapon() override {
        return new Hammer();
    }
};

```

5. Client Code

```

Hero* h = new Thor();
Weapon* w = h->createWeapon();
w->use();

```